Write Inequalities

Lesson 7-4

Name: Date: Class:

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x > 5$ means x can be 6, 7, 8, ... — any number greater than 5

Inequality

A math sentence that compares two sides with <, >, ≤, or ≥.

$8 > 3$ — eight is larger than three

Greater than

The > sign, showing one number is bigger.

$2 < 7$ — two is smaller than seven

Less than

The < sign, showing one number is smaller.

'At least 10 players' → $p \geq 10$ (could be 10, 11, 12, ...); 'At most 5 tries' → $t \leq 5$ (could be 5, 4, 3, ...)

At least / At most

'At least' means ≥. 'At most' means ≤.

In $x > 5$, the letter x stands for any number greater than 5.

Variable

A letter that stands for an unknown number.

"No more than 8 people" means people ≤ 8.

No more than

A phrase meaning less than or equal to (≤).

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can write inequalities to represent real-world situations.

1️⃣ Notice

Detective Navarro is narrowing down suspects.

2️⃣ Key idea

I can write inequalities to represent real-world situations.

3️⃣ Words I'll use

- **Inequality**
- **Greater than**
- **Less than**
- **At least / At most**
- **Variable**
- **No more than**

🤔 Think About It

- What does 'at least 18' mean about the suspect's age?
- Could the suspect be exactly 18?
- How is an inequality different from an equation?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which inequality represents: 'The temperature is less than 40 degrees'?

A. $t < 40$

B. $t > 40$

C. $t \leq 40$

D. $t = 40$

Step 1 'Less than' means <, so the inequality is $t < 40$.

✅ **Answer:** A. $t < 40$

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which inequality represents: 'You must be at least 48 inches tall to ride'?

A. $h \geq 48$

B. $h > 48$

C. $h < 48$

D. $h \leq 48$

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which inequality represents: 'A store can hold no more than 75 people'?

A. $p \leq 75$

B. $p < 75$

C. $p \geq 75$

D. $p > 75$

Show your work:

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A witness says the suspect is 'at least 18 years old,' written as $a \geq 18$. Why does the detective use an inequality here instead of an equation like $a = 18$?

Level 1 support

Start here: The age could be 18 or any number greater, so it is not a single value.

💡 **Need a hint?**

Sentence starters (optional)

An inequality fits better than an equation because ____.

Una desigualdad encaja mejor que una ecuación porque ____.

'At least 18' means the age can be ____.

'Al menos 18' significa que la edad puede ser ____.

Word bank: inequality | greater than | less than | at least / at most

Listen for: *Students explain that 'at least 18' allows many values (18 and up), so an inequality with ≥ captures it while an equation would force one value.*

Level 2

How would the meaning change if the clue used $a > 18$ instead of $a \geq 18$? Justify whether an 18-year-old still fits.

- With $a > 18$ an 18-year-old would ____ because ____.
- The line under the symbol means ____.

EXPLORE

How do the phrases 'at least' and 'at most' change which inequality symbol you choose compared to 'more than' and 'fewer than'?

Level 1 support

Start here: 'At least' and 'at most' include the number; 'more/fewer than' do not.

💡 **Need a hint?**

Sentence starters (optional)

'At least' uses ____ because it ____ the number, but 'more than' uses ____.

'Al menos' usa ____ porque ____ el número, pero 'más que' usa ____.

My inequality is ____ because the phrase says ____.

Mi desigualdad es ____ porque la frase dice ____.

Word bank: inequality | greater than | less than | at least / at most

Listen for: *Students match 'at least' to ≥ and 'at most' to ≤ (include the value), and 'more than'/'fewer than' to > and < (exclude the value).*

Level 2

Compare 'no more than 8 exits' with 'fewer than 8 exits.' Could 8 exits be allowed in each case? Justify.

- 8 exits is allowed with ____ but not with ____ because ____.
- The difference is whether the number is ____.

CONNECT

Where do you see inequalities in real life, such as age limits or speed limits?

Level 1 support

Start here: Many real rules describe a range of allowed values, not one value.

💡 **Need a hint?**

Sentence starters (optional)

I see an inequality when ____, and I would write ____.

Veo una desigualdad cuando ____, y escribiría ____.

This is an inequality, not an equation, because ____.

Esto es una desigualdad, no una ecuación, porque ____.

Word bank: inequality | greater than | less than | at least / at most

Listen for: *Students give a real example (ride height, speed limit, ticket age) and write an inequality with the correct symbol for a range of values.*

Level 2

Critique: 'A speed limit of 25 mph means you must drive exactly 25.' Write the correct inequality and explain the real rule.

- This is wrong because a speed limit means ____.
- The correct inequality is ____ because ____.

Write About the Math

The Writing Revolution

I can explain my inequality using the words inequality, greater than, less than, and at least / at most.

1. Kernel Sentence

subject + verb

Model: Inequality is a math sentence that compares two sides with <, >, ≤, or ≥.
Desigualdad es una oración matemática que compara dos lados con <, >, ≤ o ≥.

Write a kernel sentence about inequality. Use a subject and a verb.
Escribe una oración base sobre desigualdad. Usa un sujeto y un verbo.

2. Sentence Expansion

because · but · so

Kernel: Inequality matters in math
Desigualdad importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Inequality matters in math because ____.
Desigualdad importa en matemáticas porque ____.

but
pero

Inequality matters in math, but ____.
Desigualdad importa en matemáticas, pero ____.

so
entonces

Inequality matters in math, so ____.
Desigualdad importa en matemáticas, entonces ____.

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about inequality.
Di un hecho verdadero sobre inequality.

Inequality ____.

Question
Pregunta

Ask a question about inequality.
Haz una pregunta sobre inequality.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about inequality.
Muestra entusiasmo sobre inequality.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with inequality.
Dile a un compañero qué hacer con inequality.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

An inequality fits better than an equation because ____.
Una desigualdad encaja mejor que una ecuación porque ____.

'At least 18' means the age can be ____.
'Al menos 18' significa que la edad puede ser ____.

'At least' uses ____ because it ____ the number, but 'more than' uses ____.
'Al menos' usa ____ porque ____ el número, pero 'más que' usa ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A bus holds at most 40 people. Which inequality shows the number p of people allowed?

A. $p \le 40$

B. $p \ge 40$

C. $p > 40$

D. $p < 40$

Show your work:

2. Which inequality means 'a number n is less than 8'?

A. $n < 8$

B. $n \le 8$

C. $n > 8$

D. $n \ge 8$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write two different real-world situations: one that uses > and one that uses ≥. Explain why the situations need different symbols even though both mean 'bigger.'

Sentence starter: Situation 1 (>): ____. This uses > because ____. Situation 2 (≥): ____. This uses ≥ because ____. The difference is that ≥ includes ____ while > does not.

Show your work:

____

____

____

____

Reflect — Exit Ticket

Which inequality represents: 'A number n is at least 14'?

A. $n \geq 14$

B. $n > 14$

C. $n \leq 14$

D. $n < 14$

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **We Try (notes):** A. $h \ge 48$ *— 'At least' means greater than or equal to, so $h \ge 48$.*
2. **You Try (notes):** A. $p \le 75$ *— 'No more than' means less than or equal to, so $p \le 75$.*
3. **Try It 1:** A. $p \le 40$ *— 'At most 40' means 40 or fewer: $p \le 40$.*
4. **Try It 2:** A. $n < 8$ *— 'Less than 8' is $n < 8$.*
5. **Exit Ticket:** A. $n \ge 14$ *— 'At least 14' means 14 or greater, so $n \ge 14$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Inequality is a math sentence that compares two sides with $<$, $>$, $\le$, or $\ge$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).